Voluntary liquidity buffers under finite common coverage

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THE PUZZLE

Same reserve rate. Different exposure and access.

If protection is available later, how much self-insurance is worth holding now?

A buffer is chosen before liquidity is redistributed

ONE CURRENCY: USD / TWO BANK TYPES / NU RESERVES

$t = 0$ **Prepare**
Choose voluntary buffer u_k before observing either shock.

→

$t = 1$ **Allocate**
Signed flow → prescribed interbank trade → finite common coverage.

→

$t = 2$ **Settle**
Account for financing and uncovered needs.

No personal reserve reuse
Required reserves are not individually usable (NU).

Prescribed allocation
Interbank priority and i are given, not an equilibrium.

Finite, free coverage
The common facility covers part of the residual deficit.

Joint thesis architecture: Carlos Gómez Puicán and Fabián Méndez Suárez, under Marco Ortiz, incorporating his benchmark note. Individual course contribution: buffer optimization, verification and simulations. Open choices are not advisor-approved.

One private choice within a wider thesis

THE WIDER THESIS

Who receives coverage?

Exposure, access and self-insurance.

$$I_k = \frac{C_k/d_k}{\bar{C}/D}$$

Relative coverage; not a net transfer or welfare measure.

THIS INDIVIDUAL COURSE PROJECT

How much buffer should one bank choose?

Interbank access and finite coverage shape the marginal value of liquidity.

Today's route: a conditional best response first.

01 1–4	**02** 5–8	**03** 9–11	**04** 12–13
Set up the model	**Choose buffers**	**Vary capacity**	**Check evidence**

One signed flow creates deficits and surpluses

$$\omega_k = \beta_k A + \varepsilon_k, \qquad s_k = d_k(\omega_k + u_k)$$

Negative position: deficit

$D_k = \mathbb{E}[(-s_k)^+ \mid A]$

Positive position: surplus

$X_k = \mathbb{E}[(s_k)^+ \mid A]$

D_k, X_k are currency **amounts**; u_k is a **ratio to deposits**.

$$\underbrace{\sum_k \mu_k \psi_k D_k}_{\text{borrowed}} = \underbrace{\sum_k \mu_k \ell_k X_k}_{\text{lent}} = \min\{P, Q\}$$

Interbank trade redistributes liquidity. It does not create it.

Prescribed fractions: ψ_k for borrowing, ℓ_k for lending; access $\tau_k \in [0,1]$. Effective supply P and demand Q are defined in B1. If either side is zero, transacted volume and both fractions are zero.

Holding liquidity has a cost and three payoffs

PRIVATE EXPECTED NET COST PER DEPOSIT

$$\min_{u_k \in \mathcal{U}_k} \left\{ r u_k + \mathbb{E}_A \left[i \psi_k \frac{D_k}{d_k} - i \ell_k \frac{X_k}{d_k} + \kappa (1-\varphi)(1-\psi_k) \frac{D_k}{d_k} \right] \right\}.$$

01 Carry
Pay for liquidity held before shocks.

02 Borrow
Pay interest on the financed deficit.

03 Lend
Earn income on the surplus actually lent.

04 Uncovered
Pay for the final residual shortfall.

The bank knows state-contingent schedules, not realized shocks.

Frozen during an individual deviation: $\psi_k(A), \ell_k(A), \varphi(A)$ and prices. $\kappa > i \geq 0$; common coverage is free. Compact course choice domain. This is an atomistic private problem, not aggregate equilibrium.

A buffer also creates lending income

$$\pi_k^- = \mathbb{P}(\omega_k + u_k < 0 \mid A), \qquad \pi_k^+ = \mathbb{P}(\omega_k + u_k > 0 \mid A)$$

Deficit margin

$$\frac{\partial(D_k/d_k)}{\partial u_k} = -\pi_k^-$$

Surplus margin

$$\frac{\partial(X_k/d_k)}{\partial u_k} = +\pi_k^+$$

$$r = \mathbb{E}_A\Big[\Big(i\psi_k + \kappa(1-\varphi)(1-\psi_k)\Big)\pi_k^- + i\ell_k\pi_k^+\Big]$$

Marginal carrying cost = avoided deficit cost + lending opportunity.

Proposition 1: integrable signed shocks, atomless conditional law, interior choice and fixed schedules. A necessary FOC; sufficiency requires more.

Lending income can overturn convexity

$$a_k = i\psi_k + \kappa(1-\varphi)(1-\psi_k), \qquad b_k = i\ell_k$$

$$J_k''(u) = \mathbb{E}_A[(a_k - b_k) f_{\omega_k|A}(-u)].$$

A sufficient global condition

$a_k(A) \geq b_k(A)$ state by state.

Under convexity, an interior stationary point with positive local curvature is the unique global minimum.

What can fail?

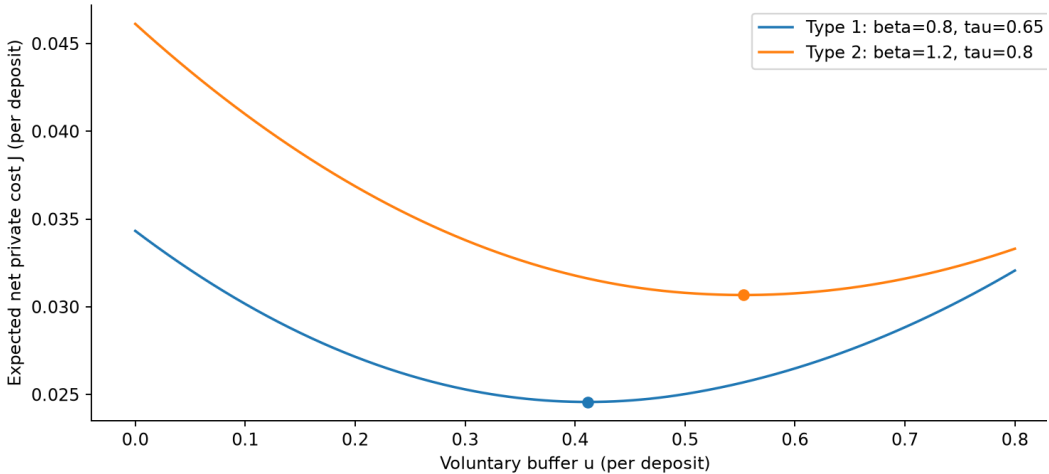
Avoided deficits add curvature; lending income subtracts it.

$\kappa > i$ alone is insufficient.

Without convexity, compare all stationary points and corners.

Density regularity required. Counterexample, corner tests and global finite-uniform solver: B5.

Two bank types choose different buffers



Type 1: $u_1^* = 0.411523$ **Type 2:** $u_2^* = 0.553381$

Large constructed buffers, not Peruvian estimates. Unique interiors; exposure/access differ. Truncated y-axis.

Common capacity weakens private self-insurance

$$\varphi(A) = \min\{1, F/\Delta(A)\}, \qquad F = \nu_F \rho D + R$$

$$\boxed{\frac{du_k^*}{dF} = -\frac{\kappa\, \mathbb{E}_A\left[1_{\{0<F<\Delta(A)\}} \frac{(1-\psi_k)\pi_k^-}{\Delta(A)}\right]}{J_k''(u_k^*)} \;\leq\; 0}$$

Regular local optimum

Interior; $J'' > 0$. Local density/IFT regularity; no coverage or uniform support kink.

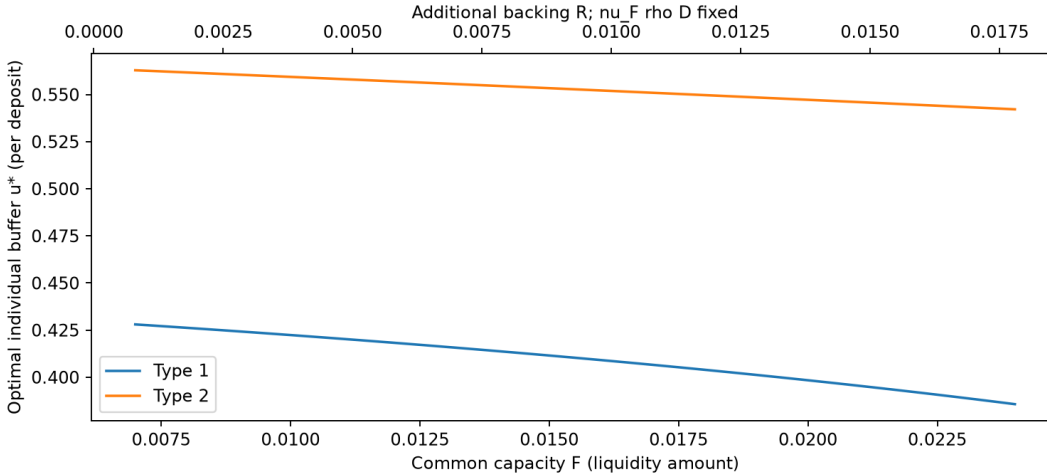
Frozen environment

$\psi_k, \ell_k, \Delta, P, Q$, prices, shock law and domain stay fixed.

Strict $<$: a positive-probability state has $0 < F < \Delta$, $\psi_k < 1$ and $\pi_k^- > 0$.

Conditional, not general equilibrium. $F > 0$; at $\Delta = 0$, use $\varphi = 0$. Saturation/no exposure: weak sign only. Global stability required; full conditions in B7.

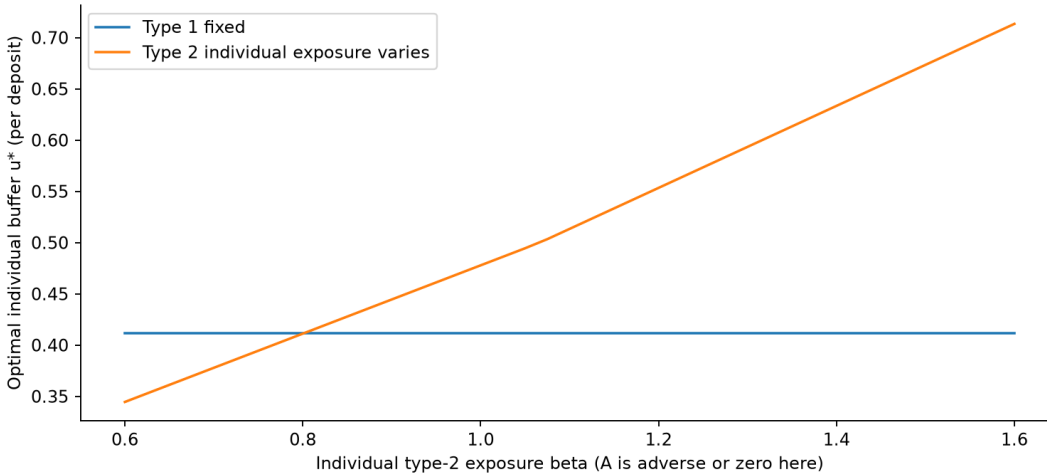
More capacity, lower voluntary buffers



At $F = 0.015$**:** $du_1^*/dF = -2.377540$ $du_2^*/dF = -1.214916$

Fixed schedules; rationed states; constructed, not GE. Truncated y-axis. Fixed-buffer I_k: invariant (B6).

Higher exposure raises buffers in this grid



$\beta_2 : 0.6 \to 1.6 \quad \Rightarrow \quad u_2^* : 0.345 \to 0.713$ **Type 1 unchanged**

One bank varies exposure; access/schedules fixed. Constructed $A \in \{-0.4, 0\}$; no universal risk theorem. Truncated y-axis.

Two kinds of evidence, with different proof scope

LEAN: AN EXISTING BUILDING BLOCK

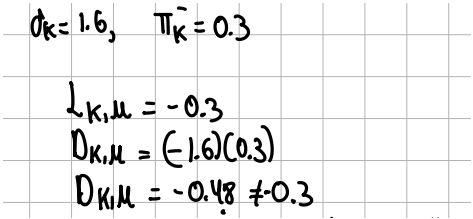
$$\widehat{L}'(t) = -\sum_j w_j \Pi_{\sigma_j}(t - m_j).$$

```
HasDerivAt (finiteLoss w sigma mean)
  (-finiteTail w sigma mean t) t := by

convert hd using 1
simp [finiteTail, Finset.sum_neg_distrib]
```

`finiteLoss_derivative`: positive widths; every component off its support knots.

HUMAN: THE STUDENT'S UNIT CHECK



Authentic manuscript 02, p. 2.

Amount: -0.48. **Ratio:** -0.30.
The amount derivative carries d_k.

Lean verifies supporting lemmas, not the new economic propositions.

Not newly Lean-verified: the post-Marco probability/reflection bridge, buffer FOC, capacity response or equilibrium. Genuine handwriting is a human check, not a Lean proof.

A conditional answer, and three open choices

Common protection can reduce private buffers.

Within the regular, frozen-schedule benchmark.

01

Allocation

Is prescribed interbank allocation an acceptable benchmark for the course/thesis transition?

02

Consistency

Is the conditional best response informative before solving the aggregate fixed point?

03

Next step

Participation incentives, shock/domain discipline, or the aggregate fixed point?

Still open: voluntary participation; aggregate consistency; illustrative shock laws and buffer domain. No general-equilibrium, welfare or optimal reserve-policy claim.

Backup material

Accounting, constructed assumptions, proof scope and model history

B1 · Full benchmark: flows and allocation

All moments and schedules below are conditional on A.

$$\omega_k = \beta_k A + \varepsilon_k, \quad s_k = d_k(\omega_k + u_k) \tag{1}$$

$$D_k = \mathbb{E}[(-s_k)^+ \mid A], \quad X_k = \mathbb{E}[(s_k)^+ \mid A] \tag{2}$$

$$\psi_k = \tau_k \min\{1, P/Q\}, \quad \ell_k = \tau_k \min\{1, Q/P\} \tag{3}$$

$$\sum_k \mu_k \psi_k D_k = \sum_k \mu_k \ell_k X_k = \min\{P, Q\}. \tag{4}$$

$$P = \sum_k \mu_k \tau_k X_k, \quad Q = \sum_k \mu_k \tau_k D_k, \quad D = \sum_k \mu_k d_k.$$

If $P = 0$ or $Q = 0$, transacted volume is zero (implemented fractions $\psi_k = \ell_k = 0$).

Current thesis candidate, analytical framework (1)–(4). Prescribed quantities are not voluntary participation conditions.

B2 · Full benchmark: protection and choice

$$F = \nu_F \rho D + R, \qquad \Delta = \sum_k \mu_k (1 - \psi_k) D_k,$$
$$\varphi = \min\{1, F/\Delta\}, \quad C_k = \mathbb{E}_A[\varphi(1 - \psi_k) D_k]. \tag{5}$$

If $\Delta = 0$, set $\varphi = 0$; common coverage has price $p_F = 0$.

$$\bar{C} = \sum_k \mu_k C_k, \qquad I_k = \frac{C_k/d_k}{\bar{C}/D}, \quad \bar{C} > 0. \tag{6}$$

$$\min_{u_k \in \mathcal{U}_k} \left\{ r u_k + \mathbb{E}_A \left[i \psi_k \frac{D_k}{d_k} - i \ell_k \frac{X_k}{d_k} + \kappa (1 - \varphi)(1 - \psi_k) \frac{D_k}{d_k} \right] \right\}. \tag{7}$$

$\nu_F \in [0, 1]$, $R \geq 0$, $\kappa > i$. NU reserves are not also individually usable liquidity. I_k is undefined if $\bar{C} = 0$.

Current thesis candidate (5)–(7). Net transfers require property, repayment and loss rules not supplied here.

B3 · Constructed course parameters

	Type 1	Type 2
Mass μ_k	0.6	0.4
Deposits d_k	1	1.6
Exposure β_k	0.8	1.2
Access τ_k	0.65	0.80
Reference u_k	0.10	0.10
Course upper bound	0.8	0.8

Common primitives	Value
Carrying cost r	0.045
Interbank rate i	0.015
Uncovered cost κ	0.25
Reserve rate ρ	0.10
Capacity share ν_F	0.05
Backing R	0.0088
Baseline F	0.015

$A = (-0.4, 0)$ with probabilities $(0.4, 0.6)$; $\varepsilon_k \mid A \sim U[-0.5, 0.5]$. Aggregate deposits $D = 1.24$. Finite states, uniforms and $[0, 0.8]$ are **course assumptions**.

Schedules come from reference buffers, not the displayed best responses.

Source: code/output/post_marco/parameters.json. All parameters constructed; no Peruvian estimates or advisor approval implied.

B4 · Conservation and fund feasibility

State	P	Q	Borrowed = lent	Fund used
Adverse	0.0189744	0.2993344	0.0189744	0.015
Normal	0.1623600	0.0721600	0.0721600	0.015

$$\sum_k \mu_k \psi_k D_k = \min\{P, Q\} = \sum_k \mu_k \ell_k X_k,$$

$$\sum_k \mu_k \varphi (1 - \psi_k) D_k = \min\{F, \Delta\} \le F.$$

Reference Δ: adverse 0.3843536; normal 0.02704.

Checks also cover $P < Q$, $P > Q$, $P = Q$, zero sides/access, $\Delta = 0$, zero capacity and saturated coverage.

These are reference-population accounting checks.

They do not make frozen-schedule best responses jointly consistent.

Sources: parameters.json, reference_market_accounting.csv, and verify.py: 99 checks across the three course models.

B5 · Convexity, corners and a counterexample

$$X_k/d_k = D_k/d_k + \mathbb{E}[\omega_k \mid A] + u$$

$$J(u) = (r - \mathbb{E}[b_k])u + \mathbb{E}[(a_k - b_k)L_k] - \mathbb{E}[b_k \mathbb{E}[\omega_k \mid A]].$$

Thus $a_k \geq b_k$ gives global convexity. Otherwise this proof fails.

Counterexample to automatic convexity:

Fully covered deficits give $\varphi = 1$ and $a_k - b_k = i(\psi_k - \ell_k)$. In the adverse reference state, $\psi_1 < \ell_1$. With $F = 10$, $J_1''(0.65) \simeq -0.003653$.

Under convexity on $[0, \bar{u}]$:

$$J'(0) \geq 0 \Rightarrow 0 \text{ minimizes}, \qquad J'(\bar{u}) \leq 0 \Rightarrow \bar{u} \text{ minimizes}.$$

Equalities can permit ties. The code compares every quadratic-region endpoint/stationary point, including nonconvex and flat cases.

Source: verify_post_marco.py; finite-uniform course specialization.

B6 · Relative incidence under fixed buffers

$$\bar{C} = \sum_k \mu_k C_k, \quad I_k = \frac{C_k/d_k}{\bar{C}/D}, \quad \sum_k \frac{\mu_k d_k}{D} I_k = 1.$$

When every state with residual deficit remains rationed and buffers and allocation stay fixed,

$$C_k = F\,\mathbb{E}_A\left[\frac{(1-\psi_k)D_k}{\Delta}\right].$$

F scales absolute coverage and cancels from relative incidence.

In this constructed sweep: $I_1 = 1.091124$, $I_2 = 0.914571$; neither changes with F.

These are **fixed-reference-buffer** values. Incidence with jointly reoptimized buffers needs aggregate consistency.

If $\bar{C} = 0$, I_k is undefined. Coverage shares do not identify net transfers, subsidy or welfare.

Source: fixed_buffer_incidence.csv. Proportional scaling does not imply directional heterogeneous incidence.

B7 · Exact conditions for the capacity result

1. Fixed deposits, domain, shock law, r, i, κ; fixed $\psi_k, \ell_k, \Delta, P, Q$. Only φ changes with F.
2. Regular differentiable interior optimum branch, $J''(u^*) > 0$; no competing global-minimum switch.
3. Valid derivative/expectation interchange. Finite-state uniforms: positive widths and no relevant support knot.
4. $F > 0$, no relevant $F = \Delta(A)$ kink. If $\Delta = 0$, use the stated zero-coverage convention.
5. Strict negativity requires positive-probability states with $F < \Delta$, $\psi_k < 1$, and $\pi_k^- > 0$. Otherwise only a weak sign is justified.

The implicit-function argument gives a **local** response. Global best-response language needs global sufficiency/stability.

Holding ν_F, ρ, D fixed gives $du^*/dR = du^*/dF$. It does not give a reserve-rate, GE or participation theorem.

POST_MARCO_AUDIT.md, section 4. Corners, ties, saturation and one-sided boundary questions are separate.

B8 · What changed across the course

	October	November 4	Current post-Marco
Core	Two currencies; public dollar backing	Direct support; asset-linked eligibility	One USD currency; prescribed interbank allocation
Buffer benefit	Avoid deficit costs	Avoid financing and deferral costs	Avoid deficit costs AND earn lending income
φ meaning	Common/public coverage	Reserve usability	Common coverage fraction
Comparative static	Conditional backing effect	Conditional reserve-burden effect	Conditional capacity effect

Historical work is preserved; its equations are not silently reused.

Existing Lean package remains pinned to paper `cf37d3f`. Current economic source: accepted post-Marco computational checkpoint.

B9 · Existing Lean proof and scope

`Gomez26EligibilityBuffers.ell_derivative` (`BufferCore.lean`), unchanged:

```
theorem ell_derivative {sigma z : ℝ} (hs : 0 < sigma)
    (hnl : z ≠ -sigma) (hnu : z ≠ sigma) :
    HasDerivAt (ell sigma) (-piTail sigma z) z := by
  by_cases hl : z ≤ -sigma
  · simpa [piTail, hl] using ell_derivative_below
      (lt_of_le_of_ne hl hnl)
  · by_cases hu : z < sigma
    · simp only [piTail, if_neg hl, if_pos hu]
      convert ell_derivative_inside hs
        (lt_of_not_ge hl) hu using 1; ring
    · simpa [piTail, hl, hu] using
        ell_derivative_above hs
          (lt_of_le_of_ne (le_of_not_gt hu) hnu.symm)
```

New deficit input: $z = u + \beta A$ (equivalently finite-loss means $m_j = -\beta A_j$).

Reflecting the symmetric uniform shock gives the probability interpretation; that bridge and the new economic propositions are not already Lean proofs.